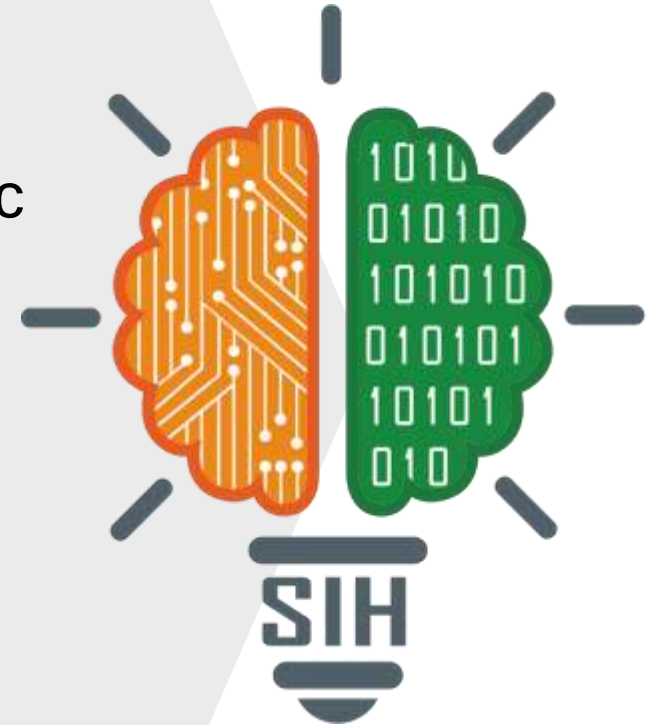


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26038
- **Problem Statement Title**- Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team ID**- SIH26-A0H-T125
- **Team Name** - Cognivex



Explainable AI for Diabetic Retinopathy Screening in Rural India

❖ Proposed Solution

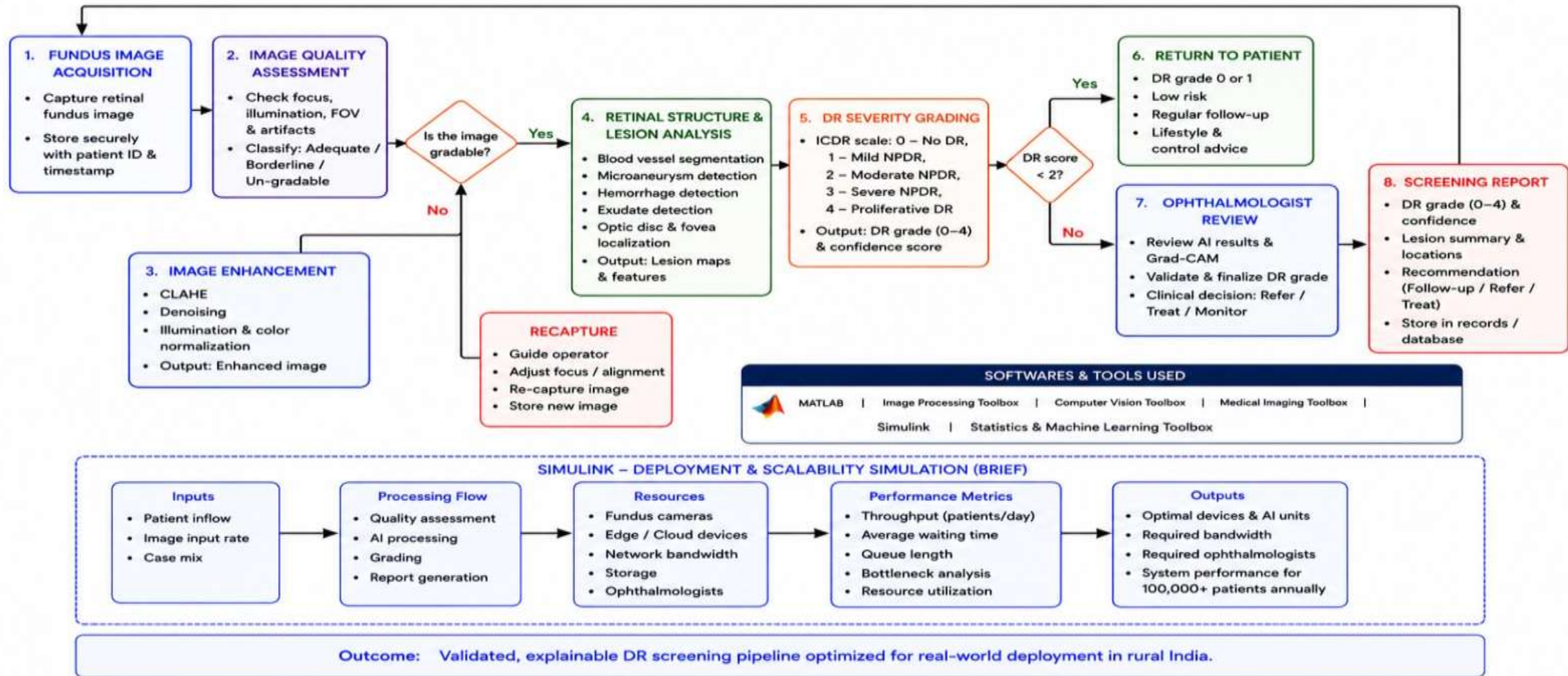
- This solution leverages Explainable Artificial Intelligence to enhance diabetic retinopathy screening in rural areas of India.
- It aims to improve early detection and diagnosis by providing transparent, interpretable AI-driven analysis.
- The system supports healthcare workers by offering clear explanations of AI decisions, increasing trust and usability.
- Designed to operate effectively in resource-limited settings, it ensures timely and accurate screening.

How the Solution Addresses the Problem

- Enhances Early Detection
- Supports Healthcare Workers
- Increases Diagnostic Accuracy
- Builds Trust through Explainability
- Operates in Low-Resource Environments

Innovation and Uniqueness of the Solution:

- Explainable AI Models Tailored for Medical Imaging
- User-Friendly Interface for Non-Specialists
- Real-Time Feedback and Reporting
- Adapted for Rural Healthcare Infrastructure
- Focus on Transparency and Accountability



FEASIBILITY	POTENTIAL CHALLENGE	STRATEGY TO SOLVE
1. Technical Feasibility - MATLAB supports image processing and deep learning. - Fundus images can be automatically analyzed. - Grad-CAM provides explainable AI results. - Simulink supports screening workflow simulation. 2. Operational Feasibility - Suitable for rural PHCs and screening camps. - AI performs initial DR screening. - Ophthalmologists review high-risk cases. - Telemedicine enables remote specialist validation. 3. Financial Feasibility - AI reduces the need for manual screening of every case. - Specialist time is focused on referable cases. - Existing/portable fundus-camera infrastructure can be utilized. - Scalable deployment reduces the need for specialist resources per patient.	1. Accuracy Challenge: Detecting tiny lesions such as microaneurysms and correctly grading DR Risk: False negatives and false positives - Missed DR / unnecessary referrals 2. Image quality Challenge: Fundus images may have blur, poor illumination, low contrast or incomplete field of view Risk: AI may produce unreliable predictions on ungradable images 3. Network Challenge: Poor rural connectivity Risk: Delayed transmission can interrupt screening and specialist consultation 4. Network Challenge: Large patient volumes with limited ophthalmologists Risk: Specialist-review bottlenecks and increased waiting time	Multi-scale lesion detection + confidence-based grading + ophthalmologist validation of high-risk/uncertain cases - Automated quality assessment → enhancement using CLAHE/normalization/denoising → recapture if still ungradable Edge/local AI processing for initial screening + compress/optimize images for transmission + store-and-forward telemedicine AI triage to prioritize referable/uncertain cases + remote specialist review + Simulink-based capacity/resource optimization

IMPACT

- Earlier identification of referable DR and reduced need for unnecessary travel for initial screening.
- AI can triage images and prioritize patients requiring specialist assessment.
- Enables retinal screening closer to where patients receive diabetes care.
- Supports tele-supported and AI-assisted screening models for larger populations.

BENEFITS

- **Social Impact** - Improves access to screening in rural areas, Enables early referral before DR progresses to vision-threatening stages. Supports remote specialist consultation through telemedicine.
- **Innovation** - Combines image-quality assessment + enhancement + lesion analysis + DR grading + Grad-CAM explainability. Provides clinical evidence alongside the AI prediction, rather than only a disease label.
Uses human-in-the-loop validation instead of replacing ophthalmologists.
- **Governance / Healthcare Management** - Produces standardized digital screening reports. Can complement PHC-based screening programmes rather than creating a completely separate healthcare pathway.

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